

Pioneer.jl Flanking-Window Features

Feature reference

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1 Motivation: wide-isolation-window DIA

In data-independent acquisition with *wide* quadrupole isolation windows, a single MS2 window co-isolates many precursors, so a confident identification needs evidence that the signal is real and not bleed-through from a co-isolated neighbor or chromatographic interference. The **flanking-window features** provide this by contrasting two regions of the *same* isolation window:

- the **core**: the contiguous run of MS2 cycles, centered on the precursor’s apex scan, where the precursor passes the search threshold (it is PSM-backed by construction — one MS2 scan per cycle per window);
- the **flank**: scans of the same isolation window in the cycles immediately *flanking* the core, where the precursor has *no* passing PSM.

A genuine precursor concentrates MS1 and fragment signal in the core with co-eluting fragments; interference smears into the flank or shows fragments that do not track the precursor.

2 Where they are computed (two pipeline stages)

1. **MainSearch** (`SearchMethods/MainSearch/scoring.jl`) — after best-per-precursor selection, `_mainsearch_flanking_core_bounds!` (:811) finds the core window, and the per-precursor loop (:1045--1097) writes the intermediate “core” columns into the fold Arrow files: `flanking_core_scan_min`, `flanking_core_scan_max`, `n_contiguous_scans`, `flanking_core_ms1_m0_signal`, `flanking_core_frag_signal`, plus the smoothed apex columns `frag1..8_smoothed_intensity`.
2. **PrecursorScoringSearch** (`PrecursorScoringSearch.jl`:199 → `add_wide_window_features_to_fold_files` in `wide_window_features.jl`) — reads the core columns, gathers the flank scans, extracts MS1 + fragment intensities from the raw spectra, computes the 11 flank feature values in `_wide_flank_feature_values` (:58), and writes them back. The two `*_signal` core columns are intermediate and dropped afterward (:893).

The final list `FLANKING_WINDOW_FEATURES` (`model_config.jl`:49) feeds the second-pass `ScoringSearch` `LightGBM`.

3 Defining the core, and what is raw vs smoothed (MainSearch)

`_mainsearch_flanking_core_bounds!` sorts a precursor’s scored rows by (`cycle`, `scan`), finds the apex (selected) scan, and walks outward while each neighbor both passes and is in the immediately adjacent cycle (± 1):

```
while lo>1 && pass_mask[prev] && cycle[prev]==cycle[cur]-1; lo-=1; end # left
while hi<len && pass_mask[next] && cycle[next]==cycle[cur]+1; hi+=1; end # right
```

It returns the core scan range `scan_min/scan_max`, `n_contiguous_scans`, and the apex/left/right boundary rows.

Core *signals* are summed from RAW intensities. Over the core scans (`scoring.jl:1082--1093`):

$$\text{core_ms1_m0_signal} = \sum_{\text{core scans}} \text{ms1_m0}, \quad \text{core_frag_signal} = \sum_{\text{core scans}} \sum_{r=1}^8 \text{frag}_r^{\text{raw}}.$$

Here $\text{frag}_r^{\text{raw}}$ are the matched per-scan intensities (`MAINSEARCH_FRAGMENT_INTENSITY_COLUMNS`); **no smoothing is applied to the sum.**

The smoothed values are a separate, per-fragment APEX height. `_mainsearch_radius1_smooth_fragme (:917)` computes, for each of the 8 fragments, a radius-1 weighted average over the apex scan and its two adjacent core scans:

$$\text{frag}_r^{\text{smoothed}} = \frac{2a_r + l_r + r_r}{\text{denom}},$$

where a_r, l_r, r_r are the raw apex/left/right intensities and `denom` counts the present neighbors. This is *not* summed — it is a denoised *peak height*. A single-scan raw apex is spiky; the 3-point weighted average is a stabler estimate of the true peak. These `frag*_smoothed_intensity` values are used (i) directly as LightGBM features and (ii) as the core apex in the flanking `frag_apex_gt2x` feature below. *So smoothing benefits the apex/peak-height comparison, not the sum.*

4 Gathering the flank and extracting signal (PrecursorScoringSearch)

- `_wide_window_index_spectra (:286)` indexes every MS2 scan by its isolation window key (`center_mz, isolation_width`) — one scan per cycle.
- `_wide_group_flank_window_scans_from_core! (:420)` collects the same-window scans in cycles flanking the core (outside `[scan_min, scan_max]`, within `WIDE_WINDOW_CYCLE_MARGIN= 3` cycles). No flank scans $\Rightarrow$ zero/default feature values.
- `_wide_top_fragment_idxes / _wide_group_fragment_windows` pick the top-8 library fragments and m/z windows; scan-centric extraction (`_wide_fill_scan_centric_{ms1_m0, fragments}!`, batched at 50000 groups) fills the *raw* flank vectors `flank_ms1_m0` and `flank_fragments` (`#flank scans  $\times$  8`).

5 The features (`_wide_flank_feature_values`, line 58)

Let c_{m0}, c_f be the core MS1-m0 / fragment signals (raw sums); a_r the smoothed core apex of fragment r ; $\mathbf{m}$ the raw flank MS1-m0 vector (one per flank scan); F the raw flank fragment matrix (scan $\times$ fragment). Floor all intensities at 0; let $s_j = \sum_r F_{j,r}$, and $\rho(\cdot, \cdot)$ be Pearson correlation (`_wide_window_pcor`; 0 for < 2 points or zero variance).

<code>flanking_ms1_m0_candidate_fraction</code>	[Float32]
$\frac{c_{m0}}{c_{m0} + \sum_j \mathbf{m}_j}$ — fraction of the precursor’s MS1 monoisotopic (m0) signal sitting in the core vs. spread into flank scans. $\rightarrow 1$ for a clean precursor.	
<code>flanking_frag_candidate_fraction</code>	[Float32]
$\frac{c_f}{c_f + \sum_j s_j}$ — the same core-vs-flank fraction for summed fragment signal.	
<code>flanking_ms1_frag_sum_corr</code>	[Float32]
$\rho(\mathbf{m}, \mathbf{s})$ across flank scans — do the MS1 precursor and the fragment sum co-vary off-apex? (co-elution sanity check in the flank).	
<code>flanking_frag_corr_mean</code>	[Float32]
mean of all pairwise $\rho(F_{:,a}, F_{:,b})$ over flank scans, among fragments with signal — average fragment-fragment co-elution in the flank.	
<code>flanking_n_correlated_fragments</code>	[UInt8]
count of fragments whose leave-one-out correlation $\rho(F_{:,r}, \mathbf{s} - F_{:,r}) > 0.7$ (WIDE_WINDOW_CORR_THRESHOLD) — how many fragments track the rest.	
<code>flanking_n_correlated_fragments_bitvec_rank</code>	[UInt16]
<code>_bitvec_pattern_rank</code> of the 8-bit mask of <i>which</i> fragments exceeded 0.7 — a learned categorical encoding of the correlated-fragment <i>pattern</i> , not just the count.	
<code>flanking_frag_corr_strength</code>	[Float32]
$\sum_r \max(\min(\rho_r, 1), 0)$ over the leave-one-out correlations ρ_r — continuous total correlation strength across the 8 fragments.	
<code>flanking_frag_corr_effective_n</code>	[Float32]
$\frac{(\sum_r \rho_r^+)^2}{\sum_r (\rho_r^+)^2}$ (participation ratio of the positive leave-one-out correlations) — the effective number of co-eluting fragments.	
<code>flanking_frag_corr_best_m0</code>	[Float32]
$\rho(F_{:,b,\mathbf{m}})$ where b is the fragment with the highest mean pairwise correlation (the “anchor”) — does the best fragment track the MS1 precursor across the flank?	
<code>flanking_signal_support</code>	[Float32]
$\frac{\#\{j : \mathbf{m}_j > 0 \text{ or } s_j > 0\}}{\#\text{flank scans}}$ — fraction of flank scans carrying any signal.	

`frag_apex_gt2x_flank_bitvec_rank` [UInt16]
`_bitvec_pattern_rank` of the 8-bit mask of fragments whose smoothed *core apex* $a_r \geq 2 \times$ their flank average $\bar{F}_{:,r}$ — which fragments genuinely peak at the apex rather than being flat/interference. (This is the one feature that uses the smoothed apex.)

`n_contiguous_scans` [UInt16]
(from the core bounds, not `_wide_flank_feature_values`) the number of contiguous cycles in the core run — how many cycles the precursor persisted.

The 11 values from `_wide_flank_feature_values` plus `n_contiguous_scans` make up the 12-entry `FLANKING_WINDOW_FEATURES`.

6 Helper functions

`_mainsearch_flanking_core_bounds!` (`scoring.jl:811`) [MainSearch]
Find the contiguous-cycle core window around the apex; return `scan_min/max`, `n_scans`, and the apex/left/right boundary rows.

`_mainsearch_radius1_smooth_fragments!` (`scoring.jl:917`) [MainSearch]
Radius-1 weighted apex smoothing $(2a + l + r)/\text{denom}$ producing `frag*smoothed_intensity`.

`_wide_window_pcor` (:5) [wide_window_features.jl]
Single-pass Float32 Pearson correlation; 0 if < 2 points or zero variance.

`_wide_candidate_signal_fraction` (:32) [wide_window_features.jl]
 $\text{core}/(\text{core} + \text{flank})$, guarding divide-by-zero.

`_wide_flank_feature_values` (:58) [wide_window_features.jl]
Compute the 11 flank features from the core scalars/apex and the raw flank arrays.

`_wide_window_index_spectra` (:286) [wide_window_features.jl]
Index scans by isolation window (`center_mz`, `isolation_width`).

`_wide_group_flank_window_scans_from_core!` (:420) [wide_window_features.jl]
Collect same-window scans in cycles flanking the core (margin = 3 cycles).

`_wide_fill_scan_centric_{ms1_m0,fragments}!` (:578/:601) [wide_window_features.jl]
Read raw-spectra MS1 m0 and fragment intensities at each flank scan (scan-centric, batched).

`_bitvec_pattern_rank` [shared]
Map an 8-bit fragment pattern mask to a learned rank (file-specific table).

7 Constants

WIDE_WINDOW_CYCLE_MARGIN= 3 (flank reaches up to 3 cycles either side); WIDE_WINDOW_CORR_THRESHOLD= 0.7 (correlated-fragment cutoff); FLANKING_SCAN_CENTRIC_BATCH_SIZE= 50000 (groups per scan-extraction batch).

8 The two LGBM feature sets

Pioneer trains two gradient-boosted classifiers with different feature sets. The flanking features belong only to the second one.

8.1 MainSearch per-file LGBM — PRESCORE_FEATURES (49 features)

Trained per file on *all* (scan, precursor) PSMs, *before* best-per-precursor selection (`features.jl:144`); used to pick the best scan per precursor and for the global prescore aggregation.

```
# Core PSM / sequence (13)
fitted_manhattan_distance, irt_error, poisson, err_norm, total_ions,
missed_cleavage, y_count, weight, gof, Mox, spectrum_peak_count,
sequence_length, fitted_hellinger
# Cross-precursor at-scan (2)
weight_ratio_at_scan, weight_rank_at_scan
# MS1 (11)
ms1_m0_mass_err_ppm, ms1_weight_apex_to_m0_apex_irt, ms1_m0_intensity,
ms1_m1_intensity, ms1_m1_to_m0_ratio, ms1_m1_to_m0_pred,
ms1_isotope_dotp_m0_m1_m2, ms1_m0_m1_m2_window_fraction,
ms1_ms2_explained_delta, ms1_m0_m1_m2_window_fraction_pc,
ms1_ms2_explained_delta_pc
# Top-8 fragment trace intensities, RAW (8)
frag1_int, frag2_int, frag3_int, frag4_int,
frag5_int, frag6_int, frag7_int, frag8_int
# Fragment-chromatogram correlations (6)
frag_apex_dispersion_irt, n_correlated_fragments,
n_correlated_fragments_bitvec_rank, frag_corr_strength,
frag_corr_effective_n, frag_corr_best_m0
# Misc (4)
top3_ms2_mass_error_mean, delta_frame_peak_center, log_by_ratio_m0, n_scans
# Backported from ADVANCED (5)
prec_mz, irt_fwhm, smoothness, log2_intensity_explained, longest_y
```

8.2 ScoringSearch (Pass-1) LGBM — ADVANCED_FEATURE_SET (73 features)

Trained *after* best-per-precursor selection, cross-run (`model_config.jl:85`; both the MBR and no-MBR paths use it, `score_psms.jl:270/362`). It mirrors PRESCORE_FEATURES but (i) substitutes the raw `frag*_int` with the smoothed apex `frag*_smoothed_intensity` (the best PSM is now known), and (ii) adds the fragment-detection, cross-window, and flanking-window families that require the selected best scan.

```
# Core PSM / sequence (13) -- same as PRESCORE
fitted_manhattan_distance, irt_error, poisson, err_norm, total_ions,
missed_cleavage, y_count, weight, gof, Mox, spectrum_peak_count,
sequence_length, fitted_hellinger
# Cross-precursor at-scan (2)
weight_ratio_at_scan, weight_rank_at_scan
```

```

# MS1 (11) -- same as PRESCORE
ms1_m0_mass_err_ppm, ms1_weight_apex_to_m0_apex_irt, ms1_m0_intensity,
ms1_m1_intensity, ms1_m1_to_m0_ratio, ms1_m1_to_m0_pred,
ms1_isotope_dotp_m0_m1_m2, ms1_m0_m1_m2_window_fraction,
ms1_ms2_explained_delta, ms1_m0_m1_m2_window_fraction_pc,
ms1_ms2_explained_delta_pc
# Top-8 fragment apex, SMOOTHED (8) -- substitutes frag*_int
frag1_smoothed_intensity .. frag8_smoothed_intensity
smoothed_2d_shadow_hellinger # (1)
# Fragment-chromatogram correlations (6) -- same as PRESCORE
frag_apex_dispersion_irt, n_correlated_fragments,
n_correlated_fragments_bitvec_rank, frag_corr_strength,
frag_corr_effective_n, frag_corr_best_m0
# Fragment-detection union/intersection (4)
n_frags_detected_union, n_frags_detected_intersection,
n_frags_detected_union_bitvec_rank, n_frags_detected_intersection_bitvec_rank
# Cross-window / transmission (4)
precursor_fraction_transmitted, n_scans_other_windows,
other_window_weight_corr, other_window_apex_delta_irt
# Flanking-window features (12) -- FLANKING_WINDOW_FEATURES (this document)
flanking_ms1_m0_candidate_fraction, flanking_frag_candidate_fraction,
flanking_ms1_frag_sum_corr, flanking_frag_corr_mean,
flanking_n_correlated_fragments, flanking_n_correlated_fragments_bitvec_rank,
flanking_frag_corr_strength, flanking_frag_corr_effective_n,
flanking_frag_corr_best_m0, flanking_signal_support,
n_contiguous_scans, frag_apex_gt2x_flank_bitvec_rank
# MS1-peak / window-precursor counts (3)
ms1_m0_peak_frag_intensity_fraction, ms1_m0_peak_n_precursors,
scan_prec_mz_n_precursors
# Misc (3)
top3_ms2_mass_error_mean, delta_frame_peak_center, log_by_ratio_m0
# RT / shape / explained (6)
n_scans, prec_mz, irt_fwhm, smoothness, log2_intensity_explained, longest_y

```

(A separate second-pass MBR-FTR model adds match-between-runs features when MBR is enabled; those are out of scope here.)

9 Consumption

The features are appended to the per-fold second-pass PSM Arrow files and feed the ScoringSearch second-pass LightGBM (FLANKING_WINDOW_FEATURES). They are most informative on wide-window instruments (e.g. Sciex SWATH), where co-isolation is heaviest, and near-constant on narrow-window data.